Notes: Adda, Dustmann, and Stevens (JPE, 2017)

The Career Costs of Children

Contents

1 Big picture

- **Question.** What are the life-cycle career costs of children for women, and how much of those costs come from lost earnings during interruptions, skill depreciation, lower skill accumulation, and sorting into more child-friendly occupations?
- **Setting.** The paper studies West German women born between 1955 and 1975 who attended lower- or intermediate-track schools and then entered apprenticeship occupations. The institutional setting is useful because occupational choice happens very early, before most fertility is realized, but after women already have beliefs about family plans.
- **Core challenge.** Fertility, labor supply, occupational choice, marriage, and savings are jointly determined. Simple wage comparisons confound the direct effect of children with ex ante sorting on ability, taste for leisure, and desired fertility.
- **Main conceptual contribution.** The paper embeds fertility, labor supply, occupational choice, assets, marriage, and human capital in one dynamic life-cycle model. This lets the authors measure not only the cost of children after birth, but also the career cost incurred before birth through selection into occupations.
- **Main hypotheses.** Three channels matter:
  - children create time out of work and lower labor supply;
  - time out of work causes skill atrophy, and the atrophy rate differs by occupation and career stage;
  - women with stronger fertility preferences sort into occupations with lower wage growth, lower atrophy risk, or better child-raising amenities.
- **Main empirical design.** The authors estimate a structural dynamic model by simulated method of moments, combining German administrative data on wages and labor supply with survey data on fertility, marriage, and savings.
- **Main baseline result.** The total career cost of children is about 35.3% of the net present value of lifetime female income, evaluated at age 15 relative to a counterfactual in which a woman knows ex ante that she will remain childless.
- **Key decomposition result.** Roughly three-quarters of the overall loss comes from reduced labor supply and forgone earnings; the remaining quarter comes from lower wages, reflecting reduced human capital accumulation, skill depreciation, and occupation choice.
- **Key heterogeneity result.** Abstract occupations have the steepest wage profiles, the largest midcareer atrophy rates, and the lowest amenity value when children are present. Women in those occupations delay fertility and are more likely to remain childless.
- **Key asymmetry across the life cycle.** Skill depreciation is not constant. It is highest around midcareer, exactly when fertility is often most desirable. This makes the timing of births central to the cost calculation.
- **Main credibility result.** The model is disciplined by 763 moments drawn from multiple data sources and matches the main age profiles of wages, work, fertility, and savings. Identification

relies on dynamic moments, occupation-choice shifters, and the joint structure that links wages, fertility, and career interruptions.

- **Economic takeaway.** Children affect careers not only because mothers work less after birth, but because expected fertility changes occupational choice and savings behavior before birth. Policies that encourage fertility therefore alter much more than short-run birth timing.

2 Data and measurement (key objects)

2.1 Why Germany is a useful laboratory

- Germany tracks students into school types after grade 4. Women in this paper attend lower- or intermediate-track schools and then usually enter apprenticeship training at age 15 or 16.
- Apprenticeship occupations are highly structured, occupation-specific, and chosen early. This makes occupational choice a meaningful long-run career decision.
- Because occupation is selected before most births occur, the setting is ideal for studying how expected fertility affects early career choice.

Interpretation. The German apprenticeship system creates an unusually clean environment in which fertility expectations can shape occupation choice long before actual childbirth.

2.2 Data source and sample construction

- **IABS administrative data.** A 2% sample of all employees contributing to German social security between 1975 and 2001. It provides wages, employment spells, occupation, education, age, and apprenticeship periods.
- **GSOEP survey data.** Used to measure fertility, household formation, family background, and spousal information for women from comparable cohorts.
- **EVS survey data.** Used to compute savings rates.
- The analysis keeps West German women born between 1955 and 1975, excludes the self-employed and civil servants, and excludes everyone who ever worked in East Germany.
- The final administrative sample contains about 2.7 million observations on wages and work spells.

Interpretation. No single data set contains everything. The paper therefore combines administrative precision on careers with survey information on fertility and household finances.

2.3 Occupation groups

- The paper groups the many apprenticeship occupations into three task-based categories:
  - **Routine occupations**
  - **Abstract occupations**

- **Manual occupations**

- The grouping follows the task-based framework of Autor, Levy, and Murnane and related work for Germany.
- Abstract jobs are expected to feature faster-changing skill requirements, steeper wage growth, and more costly interruptions.

Interpretation. The occupation grouping is not cosmetic. It is the key reduced-form way the paper maps job characteristics into fertility-related career trade-offs.

2.4 Labor supply, wages, and experience

- In the model and the data, women can work full-time, work part-time, be unemployed, or be out of the labor force.
- Skills accumulate with work: one unit per year of full-time work and one-half unit per year of part-time work.
- Wages are measured precisely in the administrative data, allowing the paper to study wage profiles and wage losses around interruptions.

Interpretation. What matters is not only whether a mother works, but how much she works and how work histories translate into skills and wages.

2.5 Fertility, marriage, and savings variables

- The GSOEP provides the timing and number of births, marital status, and spouse characteristics.
- The model allows fertility, marriage, labor supply, and savings to interact.
- The EVS data show that savings rates are hump-shaped over the life cycle and rise prior to childbirth.

Interpretation. Savings are not a side issue. They help explain why women may delay fertility even when they want children.

2.6 Dependent variables and descriptive objects

The paper focuses on several core objects:

- the occupation chosen at entry into apprenticeship training;
- labor supply by age and work status;
- wages and wage losses after interruptions;
- fertility timing and completed fertility;
- household savings rates.

2.7 Descriptive facts

- About 44.8% of women choose abstract occupations, 25.0% routine occupations, and 30.3% manual occupations.
- Occupational mobility across these three groups is very low: around 98–99% of women stay in the same occupation group from one year to the next.
- Women in abstract occupations start at higher wages and have steeper wage growth.
- After 15 years of potential experience, women in abstract jobs accumulate about 1.2 more years of total work experience and about 1.6 more years of full-time work than women in routine jobs.
- A 1-year interruption reduces wages by about 0.12 log points on average; a 3-year interruption reduces wages by about 0.21 log points. These losses are largest in abstract occupations.
- Fertility differs sharply across occupations: women in abstract jobs have their first child later, are more likely to remain childless, and are less likely to have two or more children.
- Savings rates are roughly 7% at age 20, peak near 13% around age 28, and then decline gradually.

Interpretation. The descriptive evidence already shows the central trade-off: abstract occupations pay more but punish interruptions more severely, and women in those careers adjust fertility accordingly.

3 Models and empirical specifications (all equations + interpretation)

3.1 Objective and choice set

In every period, a woman chooses consumption, whether to conceive a child, labor supply, and occupation. The model begins around age 15, when she chooses a training occupation, and continues until age 80. A period is six months long.

Interpretation. The paper is not a wage regression with controls. It is a full life-cycle problem in which fertility and career plans are chosen jointly.

3.2 Ex ante heterogeneity

The model allows four sources of heterogeneity:

$$f_i = \left(f_i^P, f_i^L, f_i^C, f_i^F\right),$$

where:

- f_i^P is labor market productivity (ability),
- f_i^L is taste for leisure,
- f_i^C is taste for children,

- f_i^F captures potential infertility.

The paper restricts the heterogeneity structure to keep estimation feasible:

- ability and taste for leisure are grouped together;
- taste for children may be correlated with ability;
- infertility is assumed orthogonal to the other traits and fixed at 5% of women based on medical evidence.

Interpretation. This is how the model separates women who earn more, women who dislike work interruptions more, and women who simply want more children.

3.3 Budget constraint

The household budget constraint is

$$A_{it+1} = (1+r)A_{it} + \text{net}(GI_{it}, h_{it}, n_{it}) - c_{it}^{HH} - k\big(\text{age}_{it}^{K}, n_{it}\big)\mathbb{1}\{l_{it} \in \{FT, PT\},\, n_{it} > 0\}. \qquad (1)$$

Here:

- A_{it} is assets,
- r is the interest rate,
- GI_{it} is gross household income,
- h_{it} indicates whether a husband is present,
- n_{it} is the number of children,
- c_{it}^{HH} is household consumption,
- $k(\cdot)$ is the cost of working when children are present, including child care.

Interpretation. Children matter through the budget in two ways: they raise consumption needs and they make work more costly through child-care-type expenses.

3.4 Skill accumulation and atrophy

Skills evolve according to

$$x_{it+1} = x_{it}\, r(x_{it}, o_{it}), \qquad (2)$$

when the woman is out of work, where the atrophy factor depends on both occupation and current skill level:

$$r(x_{it}, o_{it}) = r_1(o_{it})\mathbb{1}\{x_{it} \in [0,5)\} + r_2(o_{it})\mathbb{1}\{x_{it} \in [5,7)\} + r_3(o_{it})\mathbb{1}\{x_{it} \in [7,\infty)\}. \qquad (3)$$

While working, skills rise by one unit per year in full-time work and by one-half unit per year in part-time work.

Interpretation. A year out of work is more costly in some occupations and at some career stages than in others. This nonlinear atrophy schedule is one of the paper's central innovations.

3.5 Wage equation

Female full-time log wages satisfy

$$\ln w_{it} = f_i^P + a^O(o_{it}) + a^X(o_{it})x_{it} + a^{XX}(o_{it})x_{it}^2 + \eta_{it}, \tag{4}$$

where:

- f_i^P is unobserved ability,
- o_{it} is occupation,
- x_{it} is skill,
- η_{it} is an iid wage shock.

Interpretation. Wages differ across occupations both at entry and in how strongly they reward experience-like skills. Because skill depreciates when mothers leave work, fertility affects wages dynamically.

3.6 Marriage, divorce, and husband earnings

Marriage and divorce are modeled as probabilistic processes. Husband earnings are specified as

$$earn_{it}^h = a_0^h + a_{a1}^h age_{it}^M + a_{a2}^h (age_{it}^M)^2 + \sum_j a_j^h \mathbb{1}\{o_{it} = j\} + a_P^h f_i^P + \eta_{it}^h. \tag{5}$$

Interpretation. Marriage provides insurance and extra resources, but it also depends on characteristics that are related to women's own skills and fertility preferences. The model therefore allows assortative matching in a reduced-form way.

3.7 Conception technology

If a woman decides to conceive, a child is born next period with probability

$$p(age_{it}^M, f_i^F),$$

which declines with age and reflects possible infertility.

Interpretation. Even desired fertility is risky, especially later in life. This creates a tension between postponing births for career reasons and postponing them too long.

3.8 Value function and utility

The dynamic programming problem is

$$V_t(Q_{it}) = \max_{b_{it}, c_{it}, o_{it}, l_{it}} \left\{ u(c_{it}, o_{it}, l_{it}; n_{it}, h_{it}, \mathrm{age}_{it}^K, \Upsilon_{it}, f_i) + \beta E_t V_{t+1}(Q_{it+1}) \right\}, \tag{6}$$

where the state vector is

$$Q_{it} = \left(l_{it-1}, o_{it-1}, A_{it-1}, h_{it-1}, age^M_{it}, x_{it}, n_{it}, \text{age}^K_{it}, \Upsilon_{it}, f_i\right).$$

Utility is written as

$$u_{it} = u_1(c_{it}, l_{it}; n_{it}, f^L_i) + u_2(n_{it}; f^C_i, \text{age}^K_{it}, l_{it}, o_{it}, h_{it}) + u_3(b_{it}, \Upsilon_{it}). \tag{7}$$

Interpretation.

- u_1 captures consumption, leisure, and risk aversion.
- u_2 captures the utility from children and how that depends on the age of the youngest child, labor supply, occupation, and marital status.
- u_3 captures conception-specific preference shocks.

This structure lets the model explain why mothers may leave work when children are young and why some occupations are less attractive once children arrive.

3.9 Initial occupation choice

At the beginning of the career, women choose a training occupation according to

$$o_{i0} = \arg\max_o \left[\beta^6 E_0 V_6(Q_{i6}) - \text{cost}(o, R_i, \text{Year}_i) - q_{io}\right]. \tag{8}$$

Here:

- R_i is the region of residence,
- Year_i is the labor-market entry cohort,
- q_{io} is an occupation-specific preference shock.

Interpretation. Early occupation choice is forward-looking. Women compare future career and fertility consequences, not only contemporaneous wages.

3.10 Estimation by simulated method of moments

The model is estimated by simulated method of moments:

$$\hat{\theta} = \arg\min_\theta \left[\hat{m} - m(\theta)\right]' W \left[\hat{m} - m(\theta)\right], \tag{9}$$

where:

- $\hat{m}$ is the vector of data moments,
- $m(\theta)$ is the vector of simulated moments,
- W is a diagonal weighting matrix based on the variances of the observed moments.

The model uses 763 moments covering:

- labor supply and occupational choice,
- wages and wage dynamics,
- savings,
- fertility,
- marriage.

Interpretation. The estimation works by forcing the model to reproduce life-cycle facts from multiple data sources at the same time.

3.11 How particular moments map into parameters

- Wage changes around interruptions identify the atrophy parameters.
- Wage profiles by occupation identify returns to skills and occupation-specific wage intercepts.
- Work and fertility profiles by age identify the utility of children, the utility of leisure, and occupational amenity values.
- Savings moments identify risk aversion, child-related costs, and the consumption equivalence scale.
- Fertility residuals and wage residuals help identify the distribution and correlation of unobserved ability and taste for children.

Interpretation. The structural model is useful because it turns many reduced-form correlations into one internally consistent economic system.

3.12 Model fit

- The sample is so large that global equality of all observed and simulated moments is statistically rejected.
- Economically, however, the model fits the main trends very well: labor supply, fertility timing, wage profiles, savings profiles, and dynamic wage moments.
- Locally, the equality of observed and simulated moments cannot be rejected at the 95% level for about 53% of the moments.

Interpretation. The important point is not formal perfect fit; it is that the estimated model replicates the main life-cycle patterns that the paper wants to explain.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the model

Identification comes from the joint structure of the dynamic model plus the use of many static and dynamic moments from distinct data sets. The crucial idea is that the same parameter must explain several different facts at once.

Interpretation. For example, an atrophy parameter that explains wage losses after interruptions must also be consistent with observed fertility timing, labor supply, and occupational sorting.

4.2 Occupation-choice instruments

The initial occupation equation uses local and temporary shortages of training positions as shifters of occupation-specific entry costs. In the empirical moments, the authors use interactions between region of residence at age 16 and year of birth as instruments for initial occupation.

Interpretation. These shifters move the occupation women train for without being interpreted as direct preference shifts toward fertility.

4.3 Identification of atrophy

Atrophy is identified using regressions of wage changes around interrupted work spells on:

- interruption duration,
- experience groups,
- occupation groups,
- interactions between duration and experience.

Interpretation. Reduced-form wage losses alone are not enough because interruptions are chosen. The model uses those wage-loss moments jointly with labor supply and fertility behavior to back out structural depreciation rates.

4.4 Identification of unobserved heterogeneity

- Wage residuals from regressions of log wages on experience and occupation identify the dispersion in ability.
- Fertility residuals identify the dispersion in the taste for children.
- Correlating these residuals helps identify the correlation between ability and fertility taste.

Interpretation. This is what allows the paper to separate "high-skill women have fewer children because they want fewer children" from "high-skill women have fewer children because their career paths make children more expensive."

4.5 Identification of child-related costs and savings motives

Savings moments from the EVS are essential for identifying:

- risk aversion,
- the consumption cost of children,
- child-care-related work costs.

Interpretation. Without savings data, it would be much harder to explain why births are delayed or why transfer policies can shift timing even if completed fertility moves little.

4.6 Why the estimates are credible

- The model combines administrative and survey data rather than relying on a single source.
- Occupational switching is rare, which helps justify the focus on early occupational choice.
- The paper removes regional means and aggregate time trends from moments, relying on within-country and within-cohort variation rather than raw cross sections.

4.7 What the paper can and cannot distinguish

- The paper can separate the broad components of the career cost of children: labor supply, wage effects, skill depreciation, and occupational choice.
- It cannot fully model child quality, general-equilibrium feedbacks, or all marriage-market structure.
- It assumes women do not anticipate infertility and do not learn from repeated failed conception attempts.
- It abstracts from occupation-specific human capital transfer because occupational switching is so limited in the data.

Interpretation. The paper is strongest on decomposing career costs and simulating dynamic policy responses, and weaker on equilibrium or child-investment questions.

4.8 Relation to the literature

Relative to earlier work, the paper contributes by:

- jointly modeling fertility and labor supply rather than taking one of them as exogenous;
- adding occupational choice to the fertility-career problem;
- allowing skill depreciation to vary by both occupation and career stage;
- incorporating savings and risk aversion so the model can speak to dynamic policy effects.

5 Tables and figures

TABLE 1
DESCRIPTIVE STATISTICS, BY OCCUPATION

	Routine	Abstract	Manual	Whole Sample
Initial occupation	25.0%	44.8%	30.3%	100%
Occupation of work	25.4%	52.7%	21.9%	
A				
Annual occupational transition rates:				
If in routine last year	97.9%	1.5%	.5%	
If in abstract last year	.7%	99.0%	.2%	
If in manual last year	.9%	.8%	98.3%	
B				
Log wage at age 20	3.598	3.742	3.470	3.634
	(.297)	(.301)	(.386)	(.337)
Log wage growth, at potential experience = 5 years	.0485	.0551	.0450	.0510
	(.187)	(.156)	(.196)	(.175)
Log wage growth, at potential experience = 10 years	.0181	.0240	.0152	.0208
	(.187)	(.206)	(.223)	(.206)
Log wage growth, at potential experience = 15 years	.00995	.0147	.0127	.0133
	(.206)	(.195)	(.211)	(.200)
C				
Total work experience after 15 years	11.55	12.81	12.14	12.34
	(3.273)	(2.624)	(2.880)	(2.909)
Full-time work experience after 15 years	10.32	11.92	10.86	11.29
	(3.907)	(3.348)	(3.570)	(3.617)
Part-time work experience after 15 years	1.229	.889	1.274	1.056
	(2.187)	(1.828)	(2.125)	(1.997)
D				
Total log wage loss, after interruption = 1 year	−.0968	−.147	−.105	−.121
	(.560)	(.636)	(.633)	(.613)
Total log wage loss, after interruption = 3 years	−.152	−.253	−.223	−.216
	(.604)	(.639)	(.619)	(.625)
E				
Age at first birth	27.27	28.39	25.94	27.56
	(4.138)	(3.783)	(3.517)	(3.943)
No child (%) at age 38	14.39	20.08	14.86	17.58
	(3.067)	(2.544)	(4.164)	(1.787)
One child (%) at age 38	25.00	28.92	18.92	26.15
	(3.783)	(2.879)	(4.584)	(2.063)
Two or more children (%) at age 38	60.61	51.00	66.22	56.26
	(4.269)	(3.174)	(5.536)	(2.328)

NOTE.—Occupation of work is defined conditional on working. Log wage growth is defined for all consecutive work spells after apprenticeship training. Total log wage loss after interruption is the change in log real daily earnings between return to work and the last quarter before interruption. The total wage loss has been purged of a change in occupation, in firm size (if change of firm) and changes in hours of work. Standard deviations are in parentheses

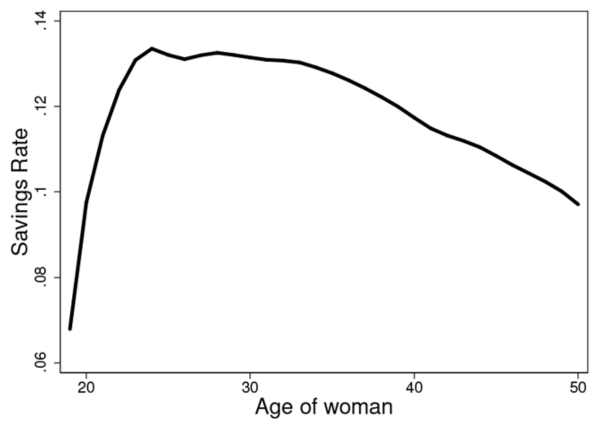

FIG. 1.—Savings rates and age: evidence from EVS data set. Computed from EVS data, by pooling the waves 1993–2008.

5.1 Table 1 and Figure 1: occupations, interruptions, and savings

Read:

- Table 1 shows that women in abstract occupations earn more initially and accumulate more work experience, but they also lose more after interruptions.
- The same table shows that fertility differs strongly by occupation: abstract-career women have their first child later and are more likely to remain childless.
- Figure 1 shows a hump-shaped savings profile: savings rise sharply in the early 20s, peak around age 28, and decline thereafter.

Economic message. The descriptive evidence already lines up with the model: women build assets before childbirth, and the costs of interruptions are not the same across career paths.

5.2 Table 2: moments used in the estimation

Read:

- Table 2 lists 763 moments used in the estimation.
- The moments span labor supply, occupation, wages, savings, fertility, and marriage.
- Several moments are dynamic, such as wage changes around interruptions and the relation between current and lagged wages.

TABLE 2
MOMENTS USED IN THE ESTIMATIONS

Moments	Data Set	No. Moments
	A. Labor Supply and Occupational Choice	
Proportion of full-time work, by age and initial occupation	IAB	25
Proportion of part-time work, by age and initial occupation	IAB	20
Proportion of out of labor force, by age and initial occupation	IAB	20
Work experience, by age	IAB	5
Annual transition rate between occupations	IAB	9
Transition rates between labor market status, by occupation	IAB	48
Proportion work, by number of children	GSOEP	15
Proportion part-time work, no child	GSOEP	5
Proportion in each occupation, initial and at all ages	IAB	6
Initial choice of occupation, by region and time period	IAB	440
	B. Wages	
Wage by age and initial occupation	IAB	21
OLS regression of log wage on experience, by occupation	IAB	9
OLS regression of log wage on age, number of children, occupation	GSOEP	12
OLS regression of log wage on past and future wages	IAB	3
OLS regression of log wage for interrupted spells on duration and experience	IAB	14
OLS regression of wage growth around interrupted work spells by occupation	IAB	10
OLS regression of husbands' log earnings on women's characteristics	GSOEP	6
Variance of residual of log wage on occupation, age, work hours	GSOEP	1
Proportion of women with log wage residual < 1 standard deviation	GSOEP	1
	C. Savings	
OLS regressions savings rate on age, occupation, number of children	EVS	24
	D. Fertility and Marriage	
Proportion with no children, by age	GSOEP	5
Proportion with one child, by age	GSOEP	5
Centiles of age at first birth	GSOEP	10
Centiles of age at second birth	GSOEP	10
Number of children at age 38	GSOEP	3
Average age at first birth, by current occupation	GSOEP	3
Proportion of childbirth within marriage	GSOEP	1
OLS regression of fertility on age and initial occupation	GSOEP	5
Instrumental variable regression of fertility on age and initial occupation (instrumented)	GSOEP	5
Mean of residual of number of children on age, by wage residual	GSOEP	2
Proportion married, by age	GSOEP	5
OLS regression marriage on age, experience, past marital status, occupation, and fertility residual	GSOEP	15
Total		763

NOTE.—IAB: Institut fuer Arbeitsmarkt-und Berufsforschung. GSOEP: German Socio-Economic Panel. EVS: Einkommens- und Verbrauchsstichprobe. Instruments for initial occupation in instrumental variable regressions are the interactions between region of residence at age 16 with year of birth.

TABLE 3
OCCUPATION-SPECIFIC PARAMETERS

Parameter	Routine	Abstract	Manual
	A. Atrophy Rates Parameters (Annual Depreciation Rates)		
At 3 years of uninterrupted work experience	−.06%	−.11%	−.03%
	(1e−5%)	(2e−5%)	(2e−5%)
At 6 years of uninterrupted work experience	−.50%	−6.90%	−3.45%
	(.11%)	(.17%)	(.24%)
At 10 years of uninterrupted work experience	−.61%	−2.65%	−3.08%
	(14.2%)	(.01%)	(.18%)
	B. Wage Equation Parameters		
Log wage constant	3.39	3.6	3.32
	(.0038)	(.0054)	(.0059)
Years of uninterrupted work experience	.1	.09	.123
	(3.3e−05)	(3.6e−05)	(.0001)
Years of uninterrupted work experience, squared	−.00382	−.0021	−.00463
	(3e−06)	(4.1e−06)	(6.4e−06)
	C. Amenity Value of Occupations		
Utility of work if children	0	−.056	−.014
		(.001)	(.0005)
Utility of part-time work if children	0	−.42	−.08
		(.003)	(.007)

NOTE.—The wage equation is defined as a function of skills—which corresponds to uninterrupted work experience—and not work experience. The former is allowed to depreciate when out of the labor force. Asymptotic standard errors are in parentheses.

Economic message. The table makes clear that the model is heavily overidentified. The authors are not fitting only one outcome; they are fitting an entire life cycle.

5.3 Table 3: atrophy, wages, and amenity values by occupation

Read:

- Table 3 shows that atrophy rates are tiny in routine jobs, much larger in abstract jobs, and often highest around six years of uninterrupted experience.
- Wage profiles are highest and least concave in abstract occupations.
- The amenity value of work and part-time work when children are present is lower in abstract occupations than in routine occupations.

Economic message. Abstract occupations are attractive before children but less child-friendly after children. That is the core occupation-fertility trade-off in the paper.

5.4 Table 4 and Figure 2: consumption costs and savings around births

TABLE 4

ESTIMATED PARAMETERS: CONSUMPTION DECISION AND COST OF CHILDREN

Parameter	Estimate
Annual discount factor	.959 (.00028)
Constant relative risk aversion utility	1.98 (.0021)
Weight of children in consumption equivalence scale	.392 (.00167)
Cost of working, if children, age ≤ 6 (€ per day)	31.1 (.36)
Cost of working, if children, age > 6 (€ per day)	12.6 (.24)

NOTE.—Asymptotic standard errors are in parentheses.

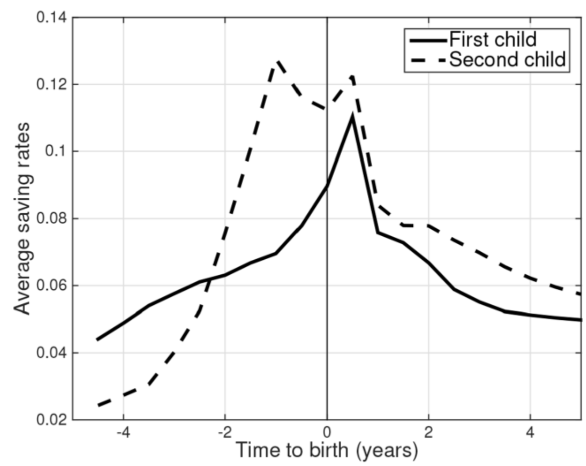

FIG. 2.—Savings rates around first and second births, model prediction. Computed through simulations of the model, involving 12,000 draws.

Read:

- Table 4 estimates an annual discount factor of about 0.959 and relative risk aversion near 2.
- The estimated child weight in the consumption equivalence scale is about 0.392.
- The cost of working when children are present is about EUR 31 per day when the youngest child is age 6 or below and about EUR 12.6 per day afterward.
- Figure 2 shows that savings rise several years before first and second births and decline afterward.

Economic message. Fertility is planned financially. Households accumulate assets before childbirth and use those assets to smooth consumption afterward.

5.5 Table 5: unobserved heterogeneity

Read:

- Table 5 estimates four types that combine low/high ability with low/high taste for children.
- High-ability women earn about 0.145 log points more than low-ability women.
- The correlation between ability and desired fertility is close to zero.
- Women with high desired fertility are more likely to sort into routine and manual occupations.

Economic message. The paper does not find that better careers are populated mainly by women who simply do not want children. Instead, fertility preferences affect occupation choice because different careers make children differently expensive.

TABLE 5
ESTIMATED PARAMETERS: UNOBSERVED ABILITY AND UTILITY OF CHILDREN

Parameter	LA/HC	LA/LC	HA/HC	HA/LC
	A. Individual Type (Ability/Fertility)			
Proportion in sample	.125	.174	.309	.393
	(8.05e–05)	(.0621)	(.00775)	(.0621)
Log wage intercept	0	0	.145	.145
	. . .	. . .	(.0026)	(.0026)
Utility of leisure	0	0	.257	.257
	. . .	. . .	(.0032)	(.0032)
Utility of one child	.484	.158	.484	.158
	(.0056)	(.014)	(.0056)	(.014)
Utility of two children	1.28	−2.04	1.28	−2.04
	(.00026)	(1.3)	(.00026)	(1.3)
Corr(ability, desired fertility)		.02		
	B. Outcome by Type			
Total fertility	1.88	.953	1.88	.951
Proporton in routine occupation	.3	.231	.301	.232
Proportion in abstract occupation	.404	.509	.407	.508
Proportion in manual occupation	.296	.26	.292	.26

NOTE.—LA: low ability; HA: high ability; LC: low taste for children; HC: high taste for children. Note that we allow ability groups to have different tastes for leisure. Asymptotic standard errors are in parentheses. Proportions in given occupation are calculated at the start of the career.

5.6 Figure 3 and Tables 6–7: career costs and the gender gap

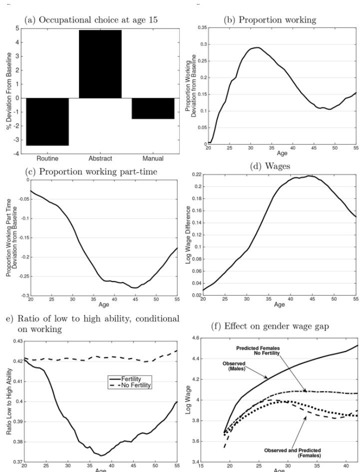

FIG. 3.—Effect of no fertility. The different panels display the difference in outcomes between a baseline scenario and one in which a woman knows that she is infertile.

TABLE 6
CAREER COST OF CHILDREN: PERCENTAGE LOSS IN NET PRESENT VALUE OF INCOME AT AGE 15, WITH AND WITHOUT FERTILITY

	Percentage Loss Compared to Baseline
Total cost	−35.3%
	A. Oaxaca Decomposition of Total Cost
Labor supply contribution	−27%
Wage contribution	−8.5%
	B. Oaxaca Decomposition of Wage Contributions
Contribution of atrophy	−1.8%
Contribution of other factors	−6.7%
Contribution of occupation	−1.6%
Contribution of other factors	−7%

NOTE.—The career costs are evaluated using simulations and comparing the estimated model with a scenario in which the woman knows ex ante that she cannot have children. The costs are computed as the net present value of female incomes, including all wages, unemployment benefits, and maternity benefits in the calculations. The discount factor is set to 0.95 annually. Initial occupation is the one in the no-fertility scenario.

TABLE 7
CAREER COST OF CHILDREN: TIMING AND SPACING OF BIRTHS

		Age at Second Birth (%)				
Age at First Birth	Only One Child (%)	22	24	26	28	30
20	−31.4	−36.4	−36.6	−36.6	−37	−36.9
22	−30.2	. . .	−34.6	−34.8	−34.8	−35.2
24	−28.1	. . .	. . .	−32.2	−32.3	−32.3
26	−26.0	. . .	. . .	. . .	−29.8	−29.8
28	−24.0	. . .	. . .	. . .	. . .	−27

NOTE.—The career costs are evaluated using simulations and comparing the scenario with no children with one in which either one or two children are born at a given age. The costs are computed as the net present value at age 15. The discount factor is set to 0.95 annually.

Read:

- Figure 3 shows the counterfactual effect of no fertility on occupation choice, labor supply, part-time work, wages, selection into employment, and the gender wage gap.
- In the no-fertility counterfactual, women are more likely to choose abstract occupations and more likely to work throughout the life cycle.
- Wages in the no-fertility scenario are about 0.22 log points higher by age 40.
- Table 6 shows that the total career cost of children is 35.3% of discounted lifetime income. About 27 percentage points come from labor supply and 8.5 percentage points from wages.
- The wage component itself includes roughly 1.8 percentage points from atrophy and about 1.6 percentage points from occupational choice.
- Table 7 shows that the first child is much more expensive than the second, and that later births reduce discounted career costs.

Economic message. The cost of children is mostly about time out of work, but wage effects are still economically important because they reflect both depreciation and early-career sorting.

5.7 Figure 4 and Table 8: policy simulation

Read:

- Figure 4 studies a transfer of EUR 6,000 at birth.
- The policy produces a short-run spike in births of about 4.5% in the first year, but the long-run increase in completed fertility is much smaller.
- The figure also shows that younger cohorts adjust savings strongly in anticipation of the policy.
- Table 8 shows that the policy matters most for women who are youngest when the policy starts: they have children earlier, work a bit less, accumulate slightly less skill, and sort modestly toward routine and manual jobs.

Economic message. Reduced-form short-run fertility responses overstate long-run effects. The policy mainly changes timing, and it changes career and savings decisions especially for women who can adjust early in life.

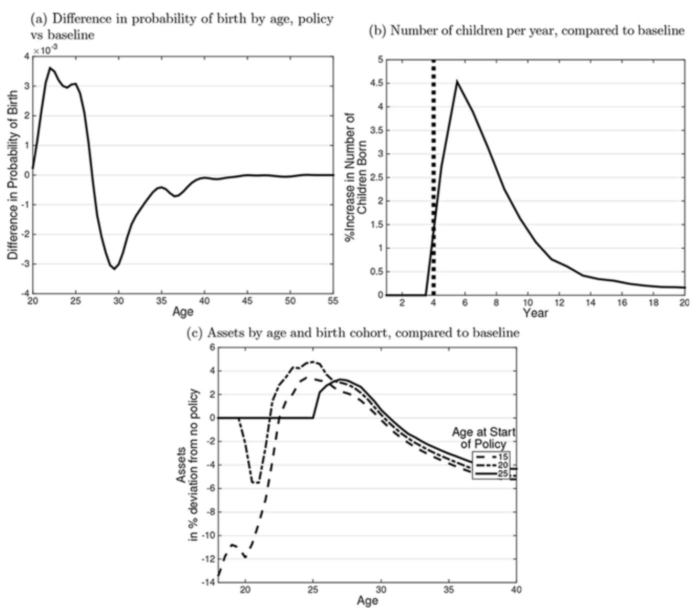

FIG. 4.—Effect of child premium. Panel a shows the effect of the policy (cash transfer of €6,000 at birth) by age on the probability of giving birth, comparing the policy to the baseline. In the policy scenario, women learn at age 15 about the policy. Panel b depicts the aggregate effect of the policy, by year, in an overlapping generation economy. The graph aggregates each year the behavior of women aged 15–60. Each year a new cohort of 15-year-olds enters the economy and the cohort who is 60 exits. The policy starts in year 4. Panel c displays the percentage change in assets as a function of age, compared to a baseline without transfer. The birth cohort who is 15 at the start of the policy can adjust right away their behavior. The cohorts who are 20 or 25 when the policy starts do not anticipate the policy.

TABLE 8
EFFECT OF INCREASED CHILD BENEFITS

	Age at Start of Policy			
	15	25	35	45
Change, no child (%)	−.8%	−.7%	0%	0%
Change, one child (%)	−.08%	−.05%	−.05%	0%
Change, two children (%)	.2%	.2%	.07%	0%
Change, age at first birth (years)	−.4	−.1	−.0005	0
Change, age at second birth (years)	−.04	−.007	.002	0
Change, skills (%)	−.29%	−.11%	−.049%	−.0019%
Change, number of years working	−.08	−.03	−.01	−.0004
Change, number of years working part-time	.04	.01	−.007	−.0003
Change, proportion routine	.3%	0%	0%	0%
Change, proportion manual	.07%	0%	0%	0%

NOTE.—The table compares two scenarios, a baseline one and one that introduces a cash transfer at birth of €6,000. Changes in fertility, skills, and work experience are computed at age 60. Changes in occupations are computed at age 15. Simulations are performed over 12,000 individuals.

6 Short summary

- The paper estimates a dynamic life-cycle model in which women choose fertility, labor supply, savings, and occupation jointly.
- Germany's apprenticeship system is crucial because occupation is chosen early and is closely tied to later career paths.
- Abstract occupations pay more and grow faster, but they also impose larger skill losses and lower amenity when children are present.
- The total career cost of children is about 35.3% of discounted lifetime income.
- About three-quarters of that cost comes from forgone labor supply; the remainder comes from wage effects due to lower skill accumulation, atrophy, and pre-birth occupational sorting.
- Fertility explains an important part of the gender wage gap, especially in women's 30s.
- A pronatalist transfer produces a large short-run fertility response but only a modest long-run completed-fertility response.
- The policy also affects occupation choice, work, skills, and savings, especially for younger cohorts.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that the career costs of children are large, dynamic, and partly determined before children are born. A woman who expects to have children changes not only labor supply after birth, but also occupation choice and savings before birth. Because occupations differ in wage growth, skill depreciation, and child-friendliness, fertility plans shape careers from the very beginning.

7.2 Economic intuition

Children change both the return to work and the value of time at home. When interruptions are costly, women may select into occupations with flatter wage profiles or better amenities for motherhood. They also accumulate savings before childbirth to finance lower income and higher expenses. The observed wage gap after children therefore reflects both post-birth behavior and earlier ex ante career planning.

7.3 Takeaway

The paper's message is broader than "children reduce wages." It shows that children reshape the entire life cycle: occupation, labor supply, experience, assets, and wage growth. That is why policy interventions aimed at fertility can have long-run career effects and why short-run policy estimates can be misleading if they ignore these dynamic adjustments.